

Origins and Evolution of Language

Week 1: Introduction

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Communication is widespread, but language is unique

How did language evolve?



Language is **universal** in our species

Language is a hugely **adaptive** trait



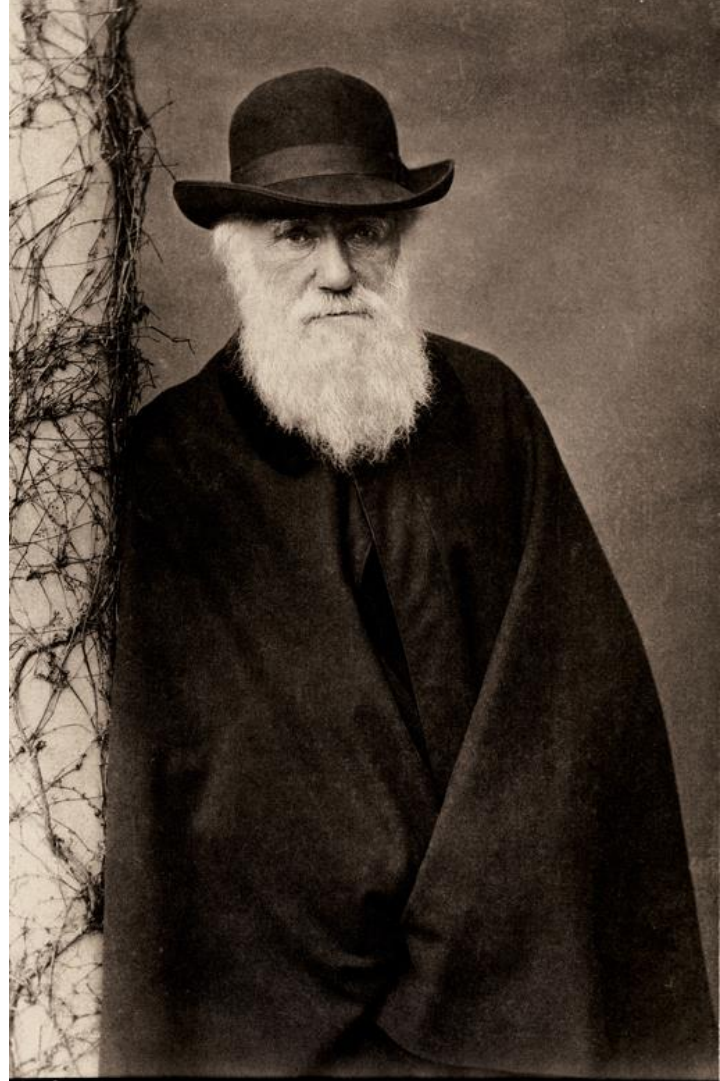
A tool for the communication of knowledge and internal states



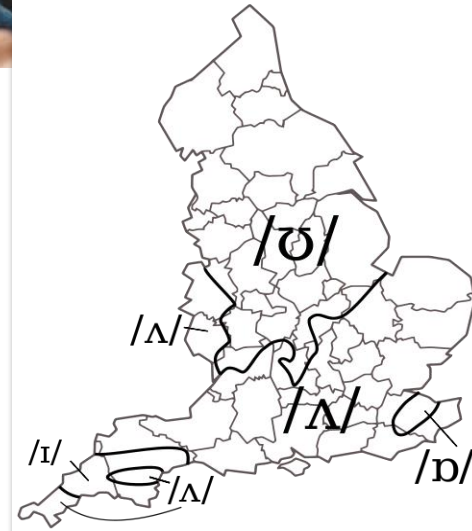
One possible explanation

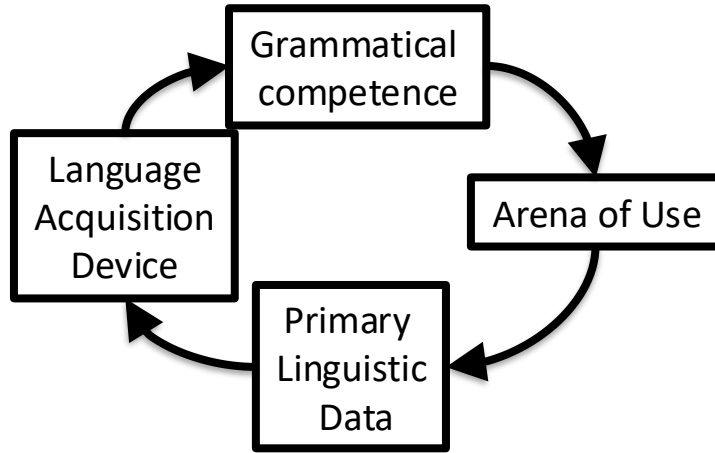
- Language is just like any other adaptive feature of an organism's biology
- It's an **innate** feature of the human mind
- It evolved by natural selection under pressure for communication

More on this next week



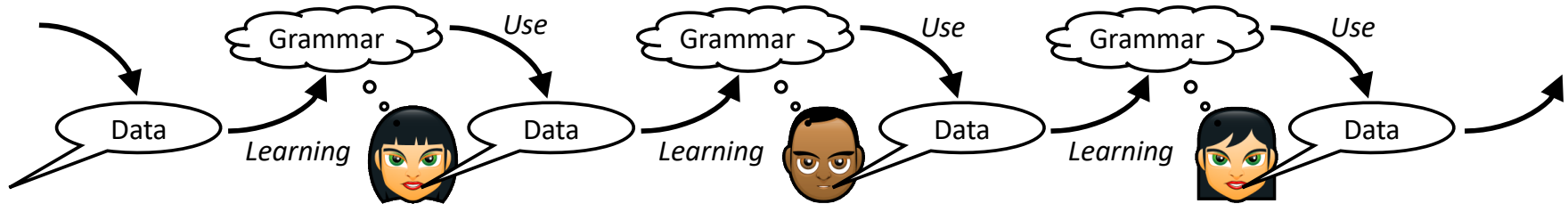
Social learning is ubiquitous in humans





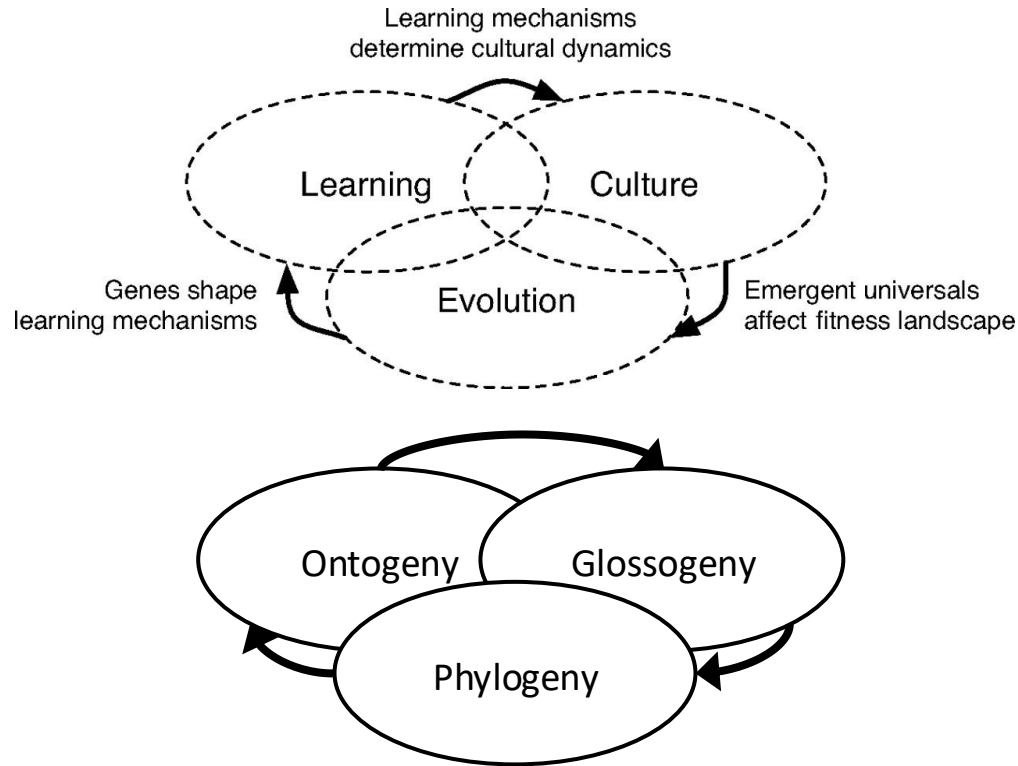
Language is transmitted
via repeated **learning**
and **use**

Language is shaped as a
consequence of these
processes



Upper: from Hurford, J. R. (1990). Nativist and functional explanations in language acquisition. In I. M. Roca (Ed.), *Logical issues in language acquisition* (pp. 85–136). Dordrecht: Foris.
Lower: from Smith, K. (2022). How language learning and language use create linguistic structure. *Current Directions in Psychological Science*, 31, 177-186.

Learning, **culture** and biology



Upper: from Kirby, S., Dowman, M., & Griffiths, T. (2007). Innateness and culture in the evolution of language. *PNAS*, 104, 5241-5245.

Lower: adapted from Fitch, W. T. (2010). *The Evolution of Language*. Cambridge: Cambridge University Press

<https://kennysmithed.github.io/origins26/>

Schedule

Week	Topic
1	Introduction
2	Natural selection, adaptation and language
3	Intention and structure in animal communication
4	<i>No class</i>
5	Social learning and cumulative culture
6	Vocal learning, grammar learning
7	Evolution of social cognition
8	Cultural evolution of language
9	Sign language and language origins
10	Gene-culture co-evolution
11	Pre-exam Q&A

Pre-lecture preparation

- **Readings must be done in advance**
- Do the reading, answer the quiz questions on Learn
 - Most useful bit for me is the free comment box at the end
- I will assume you have done the readings, we'll talk about them in class
- Each quiz includes at least one exam-style short answer question

wooclap.com for in-class voting

Go to wooclap.com

Use code INMRBS

I am:

1. An undergraduate student
2. A postgraduate student
3. Something else (e.g. here by mistake)

Workshops

- Workshops start in week 2
- First tutorial: evolution games
- Later weeks: usually paper discussion / evaluation
- **Attendance will be taken**



Prof. Simon
Kirby

+ me

Be prepared for the workshops

Assessment by exam

- Short answer questions covering
 - The lecture readings
 - The workshop readings
 - The lectures
- We'll talk more about the exam throughout the course
- The pre-lecture quiz gives you the chance to see and practice answering exam-style questions

Industrial dispute, strike action

- University management plan to reduce the University's annual budget by £140 million, £90 million from staffing (up to 2000 jobs)
- The main staff union (UCU) believe there is a capital overspending and mismanagement problem, not an over-staffing problem
- UCU want management to commit to no redundancies; management will not
- UCU members have voted for 5 weeks' action starting Monday of week 2 until management agrees to protect our jobs and your education
- **But ... going on strike is expensive, this is the latest in a long-running series of disputes, and recent strikes have not been effective in achieving our goals**
- I am not currently planning to go on strike

STUDENT-STAFF ASSEMBLY

FORCED COURSE CLOSURES

£50m of UNETHICAL INVESTMENTS

PROGRAMME CUTS

MASS STAFF CUTS

The university management has undermined and ignored staff and students for too long.

COME TOGETHER, HEAR EACH OTHER, STAND UNITED

24/09/2026 - 17:00-19:00 - APPLETON TOWER, LECTURE THEATRE 4

scan to register

Supported by:

UCU University and College Union

EUPS

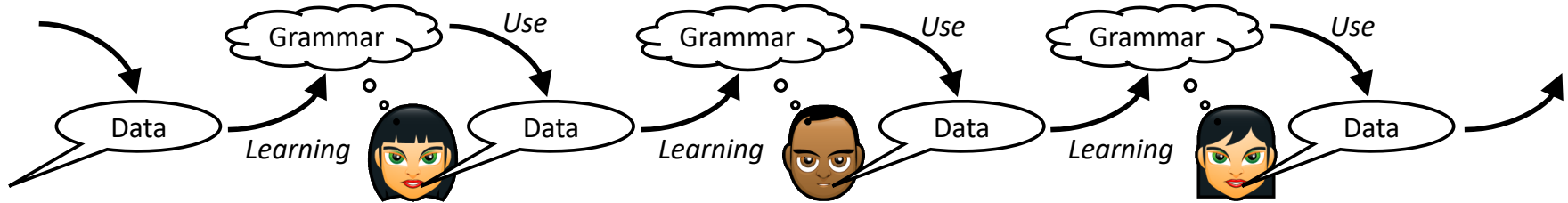
unite the UNION

UNISON

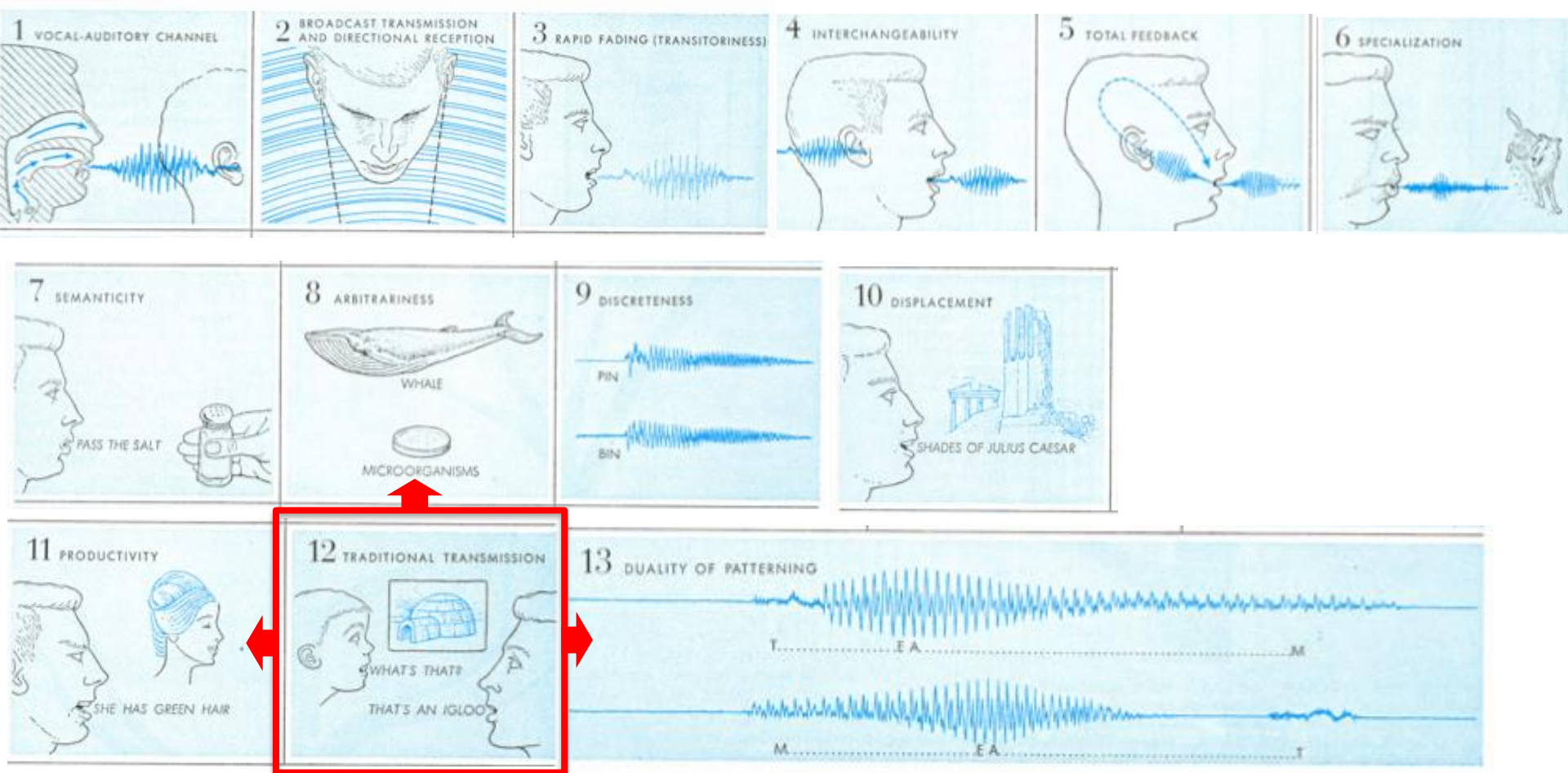
Any questions on course structure,
assessment, admin etc?

A short preview of where
we are headed

Learning, use, and language design



- Language is passed from person to person by **learning**
- People learn from language as it is **used in communication**
- Language **evolves** in response to its learning and use
- Linguistic structure is a product of this evolutionary process



Hockett's design features

Hockett, C. F. (1960). The origin of speech. *Scientific American*, 203, 88–96.

What's required for this to happen?

Social learning,
vocal learning,
grammar learning



Mitteilungsbedürfnis
and mindreading



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Social learning,
vocal learning,
grammar learning



Mitteilungsbedürfnis
and mindreading



The idea

- Humans ended up with an unusual combination of traits: ubiquitous social learning (including of vocal signalling) and deep mental interpenetration
 - We can see similar capacities in non-human animals, allowing us to understand the evolutionary history of these traits
- This set in place a cultural evolutionary process that shaped how language works
 - We can study these cultural evolutionary processes in the historical record and in the lab

What's the evidence?

- We'll look at social learning and mental interpenetration in humans **and other animals**
- We'll look at how learning and use of linguistic (or pseudo-linguistic) systems shapes their structure

A word about the bad old days of evolutionary linguistics

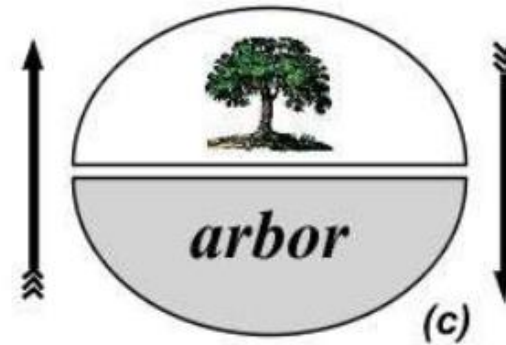
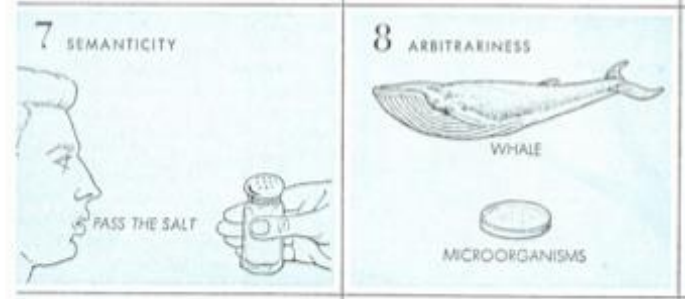
“the field has certain advantages. [...] To enter it costs little: you can’t do experiments, so no expensive equipment is required [...] It’s still a pencil-and-paper field, though with immeasurable amounts of reading and thinking involved. It is, accordingly, an ideal field for any ambitious young scholar itching to make his [sic] academic bones.” (Bickerton, 2007, *Lingua*, p. 525)

Modern evolutionary linguistics is a data-driven, empirical field
(It already was in 2007, but Bickerton was behind the times)

Some fun examples of what learning and use can do
(with a focus on Hockett's design features)

Where do symbols come from?

- **Icon:** signals resemble meanings
- **Symbol:** *arbitrary* relationship between signal and meaning



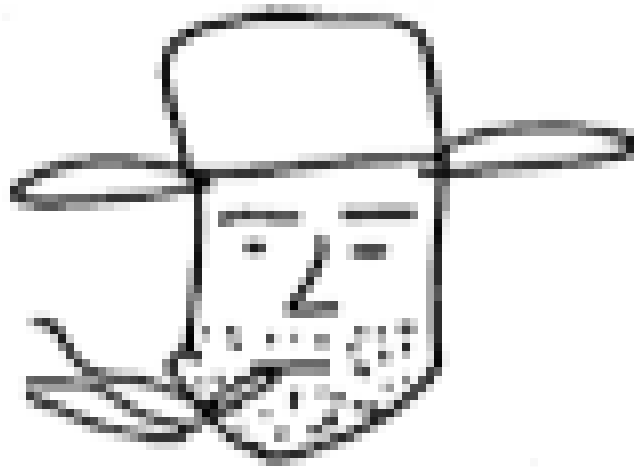
Ritualization in the lab, with humans

Repeated interaction in a Pictionary-like communication task

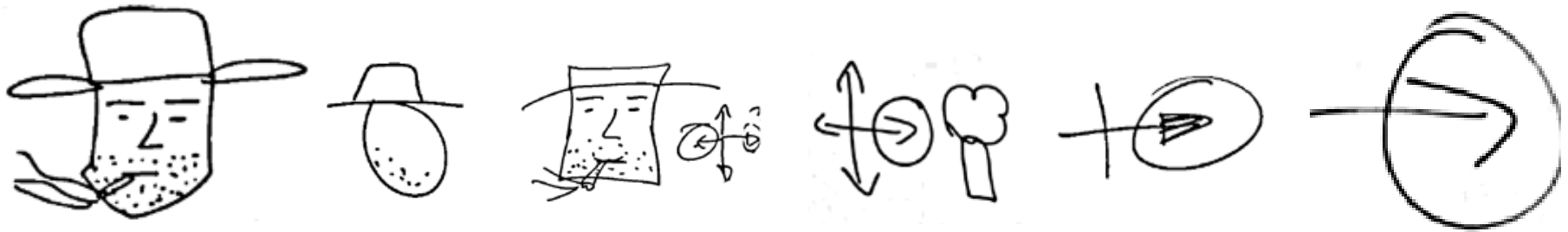


Ritualization in the lab, with humans

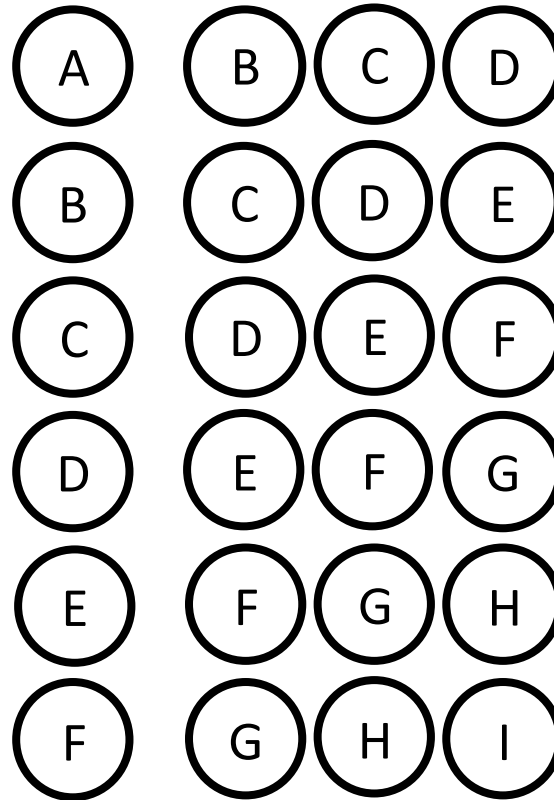
Repeated interaction in a Pictionary-like communication task



Ritualization in the lab



Transmission in laboratory 'societies'



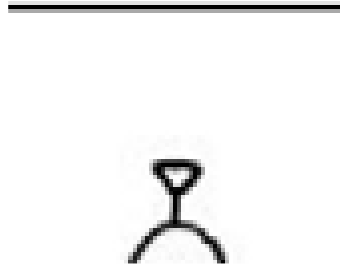
Caldwell, C. A., & Smith, K. (2012). Cultural evolution and the perpetuation of arbitrary communicative conventions in experimental microsocieties. *PLoS ONE*, 7, e43807.

Transmission in laboratory 'societies'

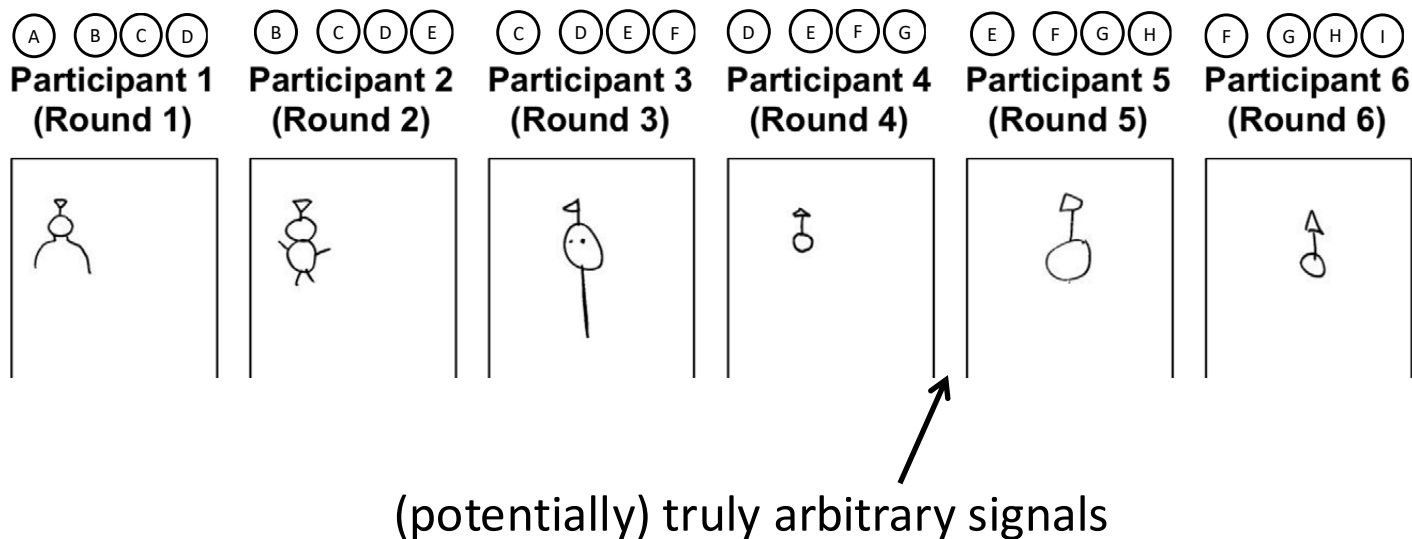
(Round 0)



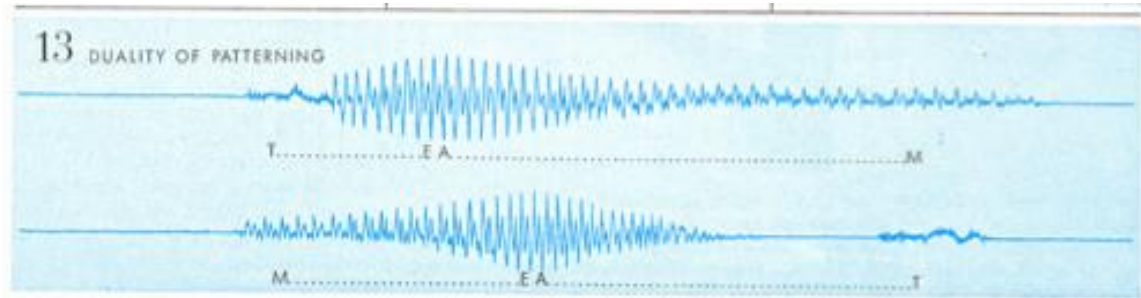
Transmission in laboratory ‘societies’ (Rou



Transmission in laboratory 'societies'



So much for symbols – how about structure?



Language's communicative power comes (in part) from its **structure**

Inventory of meaningless units
(10s)



Inventory of meaningful units
(1000s)



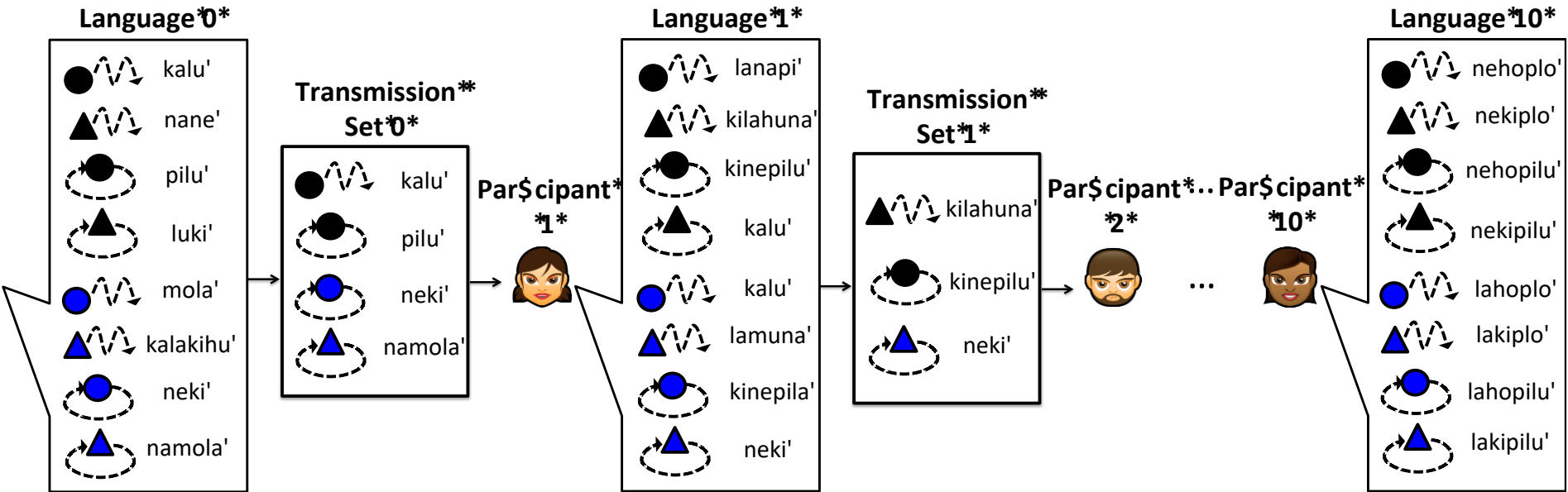
Inventory of meaningful sentences
(∞)

p t d s ɒ k g ɔ ə a ...

ə ɒə -əd dɔg kat ɒat spɔt ...
(a) (the) (past tense) (dog) (cat) (that) (spot)

the cat spotted the dog a dog spotted the cat
a cat spotted the dog the dog spotted the cat
the cat spotted the cat that spotted a dog ...
the dog spotted the cat that spotted the dog

Studying language transmission in the lab



Kirby, S., Cornish, H., & Smith, K. (2008). Cumulative cultural evolution in the laboratory: an experimental approach to the origins of structure in human language. *Proceedings of the National Academy of Sciences, USA*, 105, 10681-10686.

Figure from Kirby, S., Griffiths, T. L., & Smith, K. (2014). Iterated learning and the evolution of language. *Current Opinion in Neurobiology*, 28, 108-114.

An initial **holistic** language

--→	wimaku	miniki	gepinini	□
	nihepi	wigemi	mahekuki	○
	wikima	nipikuge	hema	△
↻	miwiniku	pinipi	kihemiwi	□
	kinimapi	wikuki	kikumi	○
	miwimi	nipi	wige	△
↻	gepihemi	kunige	miki	□
	pikuhemi	kimaki	pimikihe	○
	mihe	winige	kinimage	△

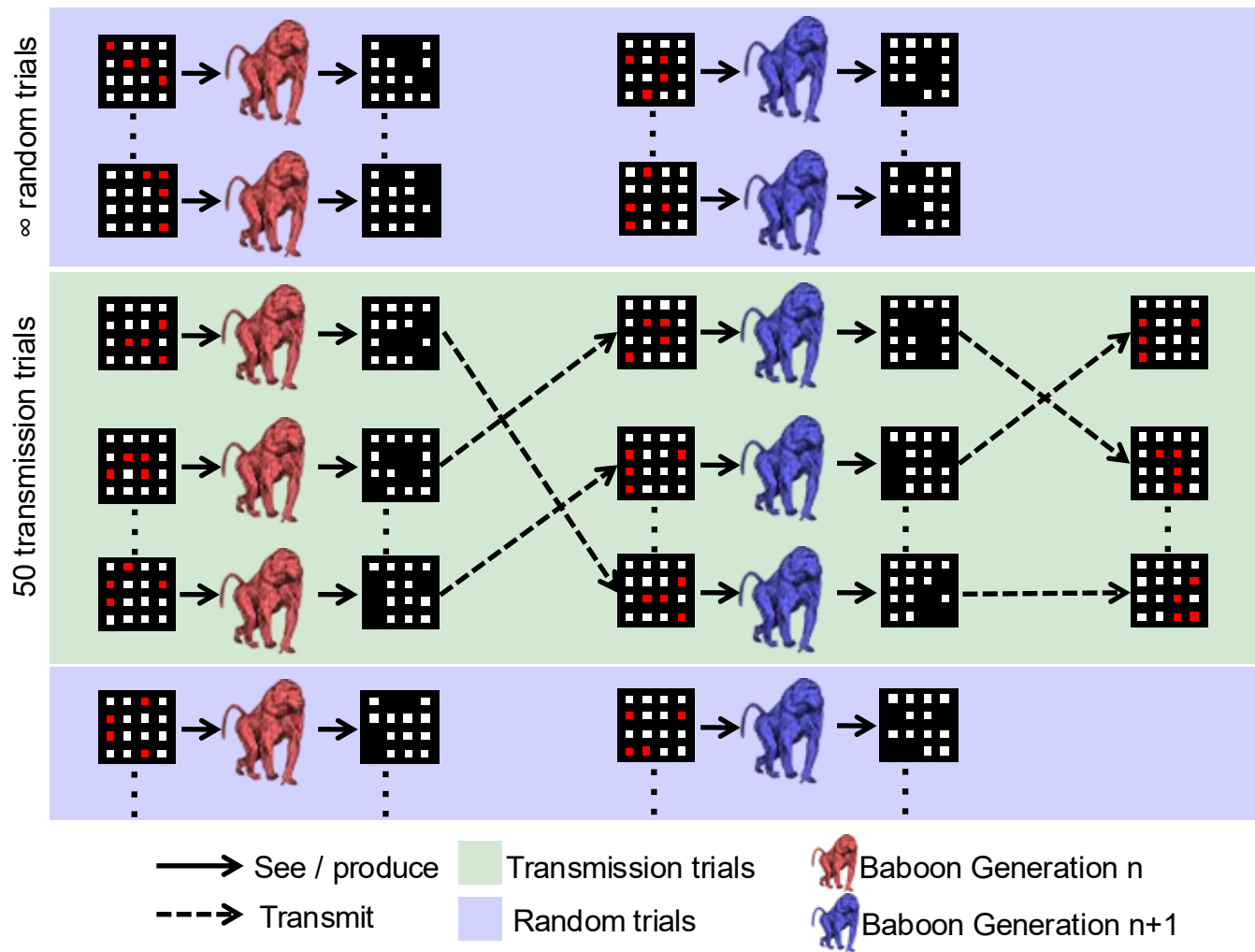
10 generations later...

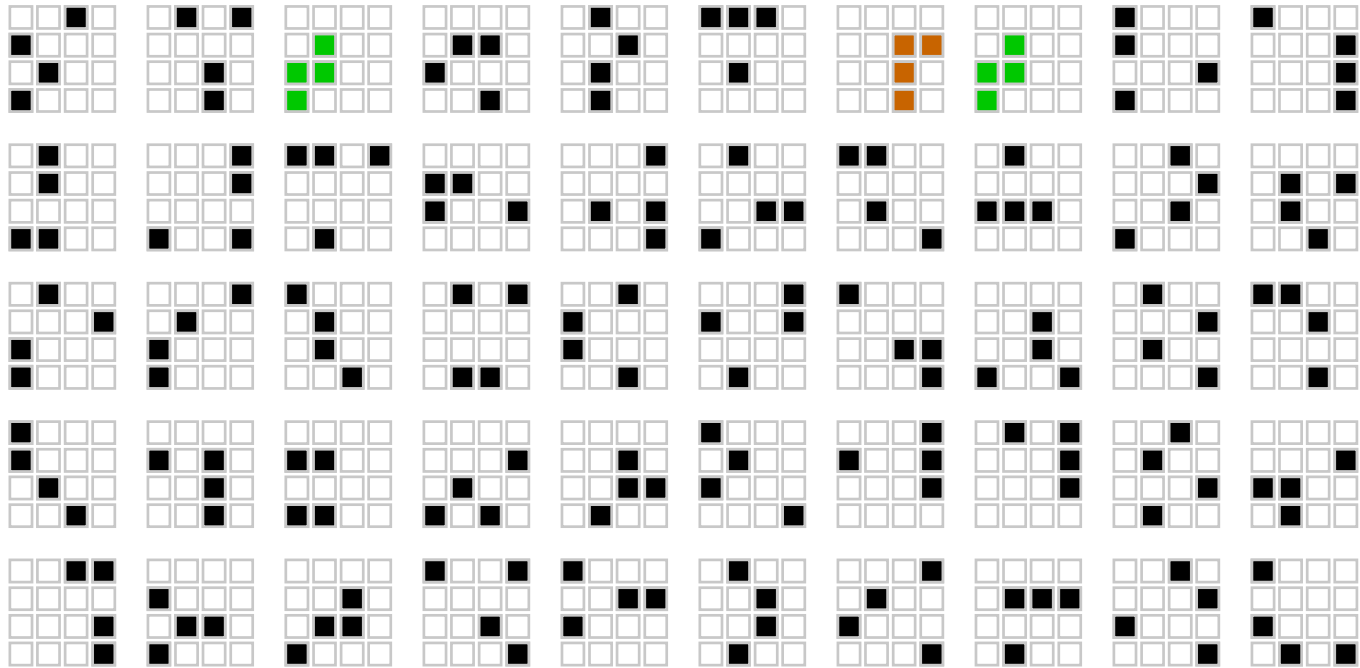
-->	ne-re-ki	le-re-ki	renana	□
	ne-he-ki	la-ho-ki	re-ne-ki	○
	ne-ke-ki	la-ke-ki	ra-he-ki	△
↻	ne-ro-plo	la-ne-plo	re--plo	□
	ne-ho-plo	la-ho-plo	re-ho-plo	○
	ne-ki-plo	la-ki-plo	ra-ho-plo	△
↻	ne--pilu	la-ne-pilu	re--pilu	□
	ne-ho-pilu	la-ho-pilu	re-he-pilu	○
	ne-ki-pilu	la-ki-pilu	ra-ho-pilu	△

If structure arises from social learning,
why isn't it more common?

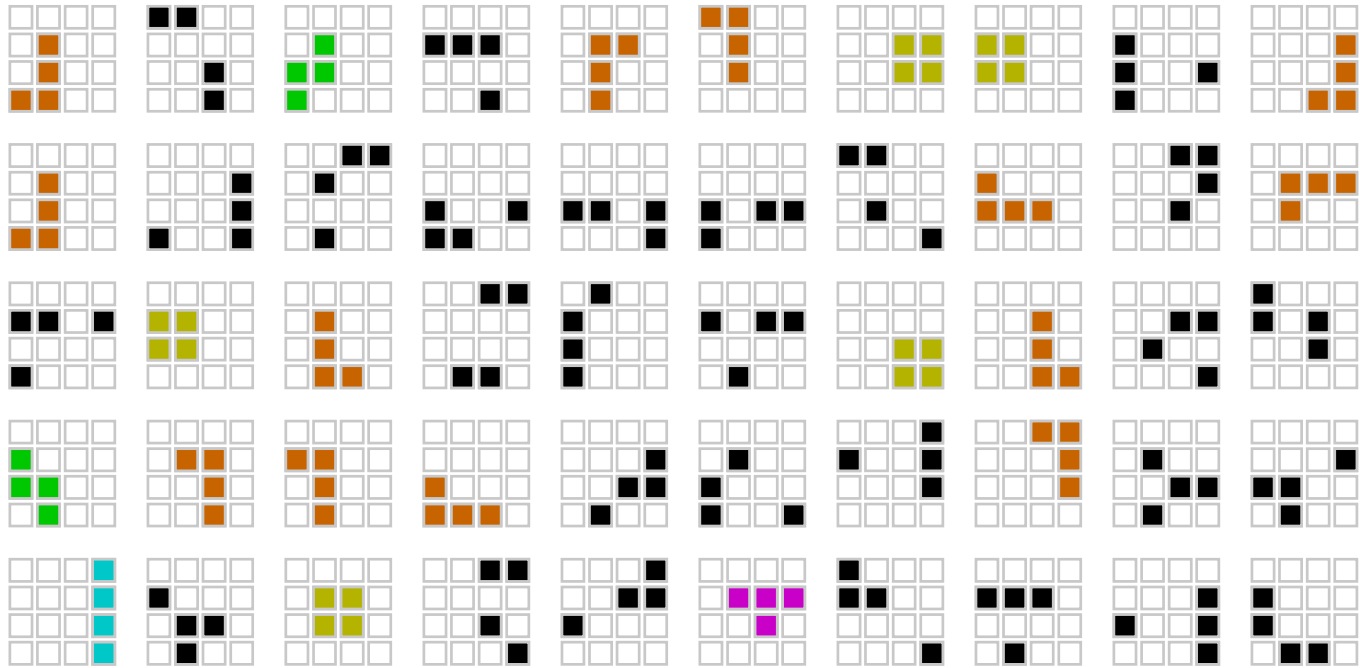




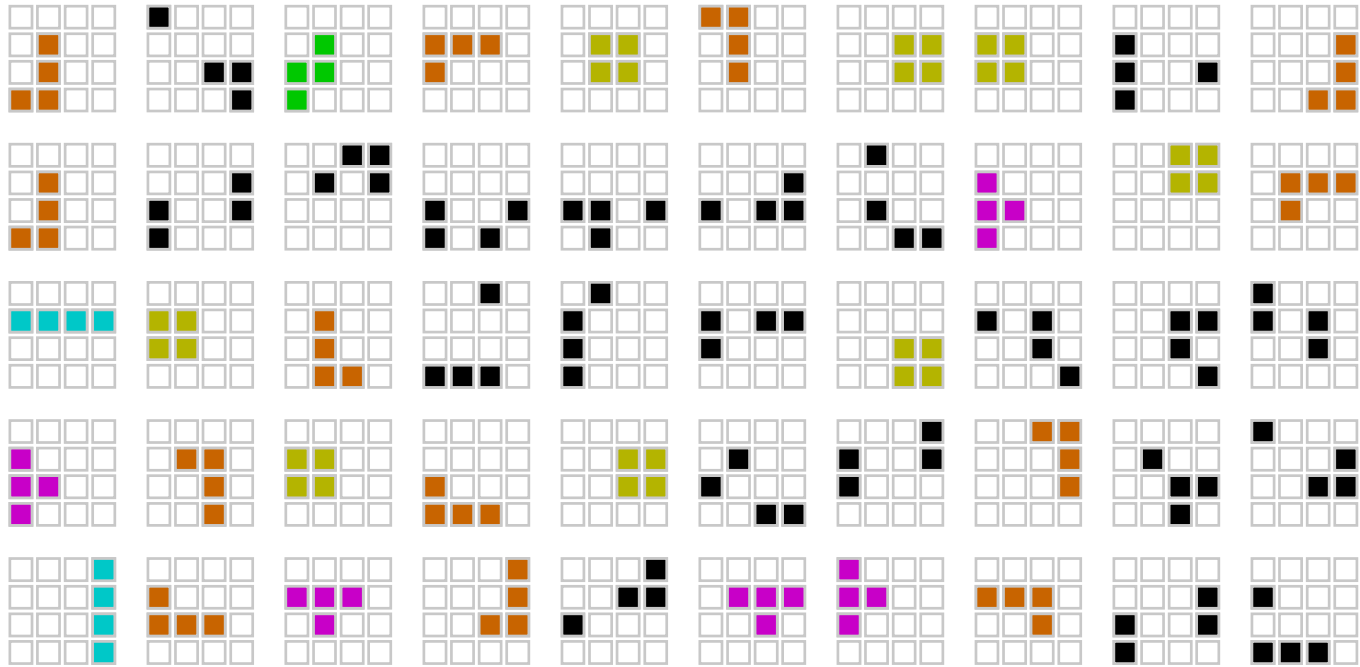




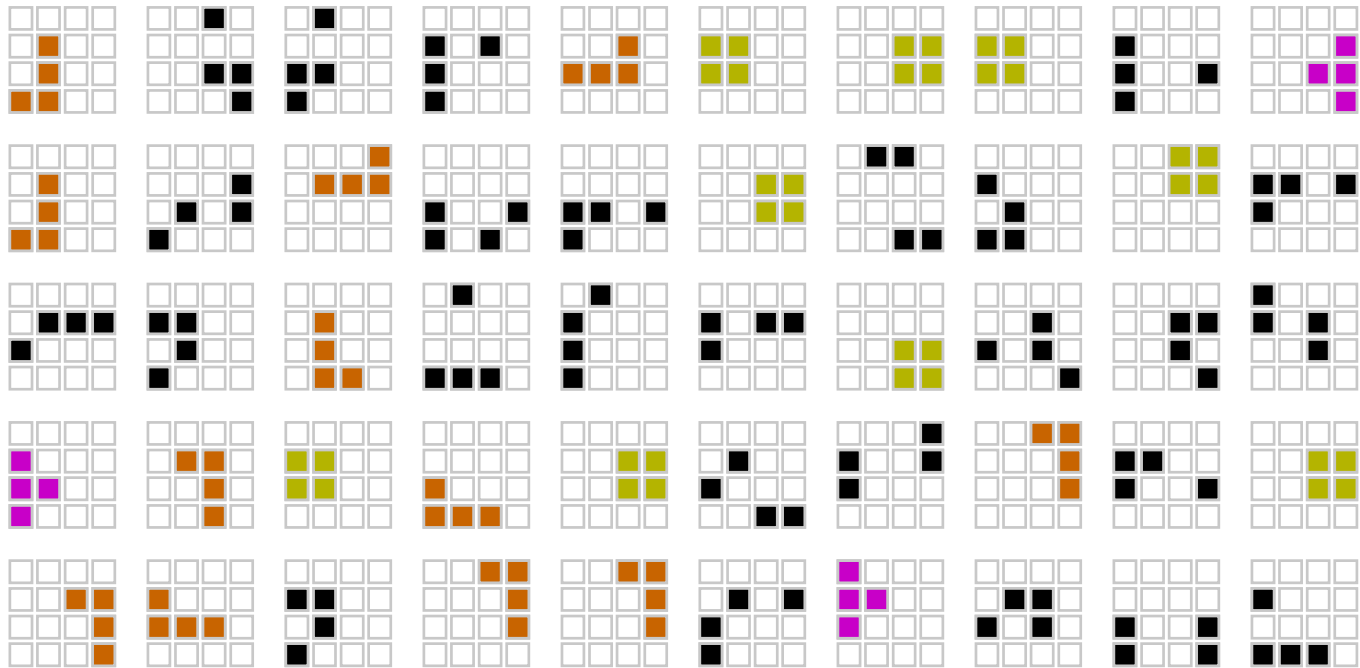
Random grids



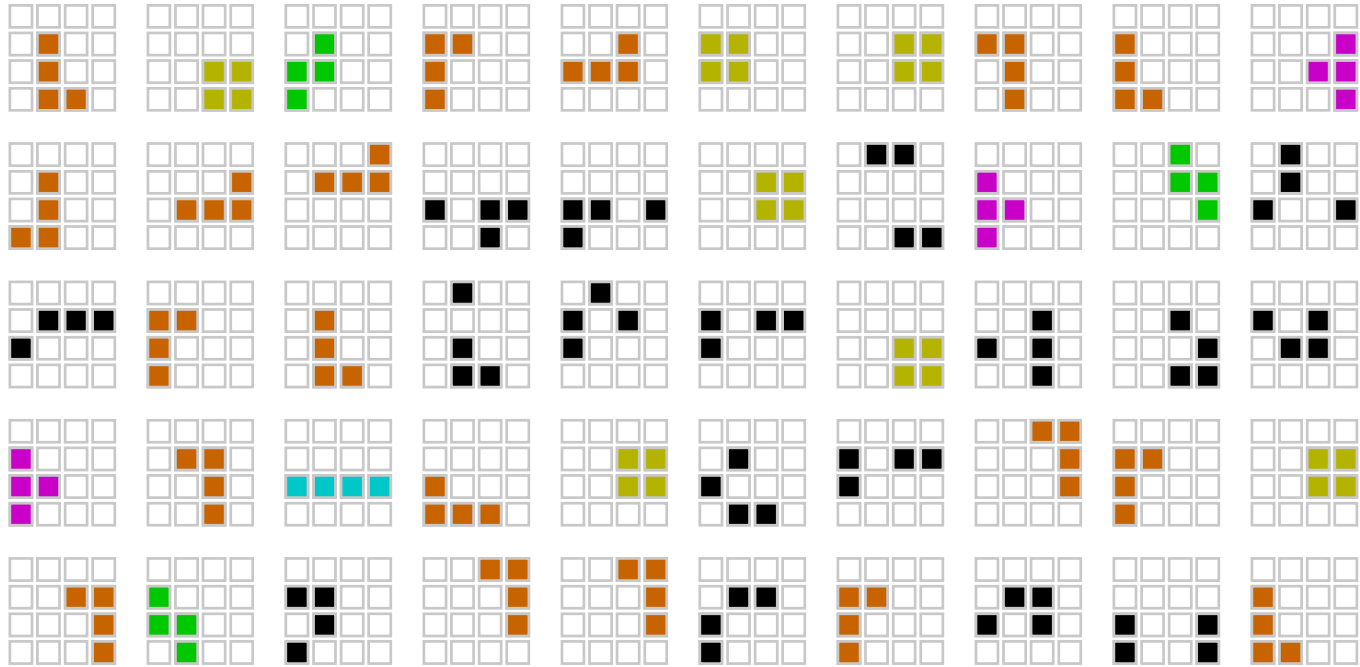
Generation 1



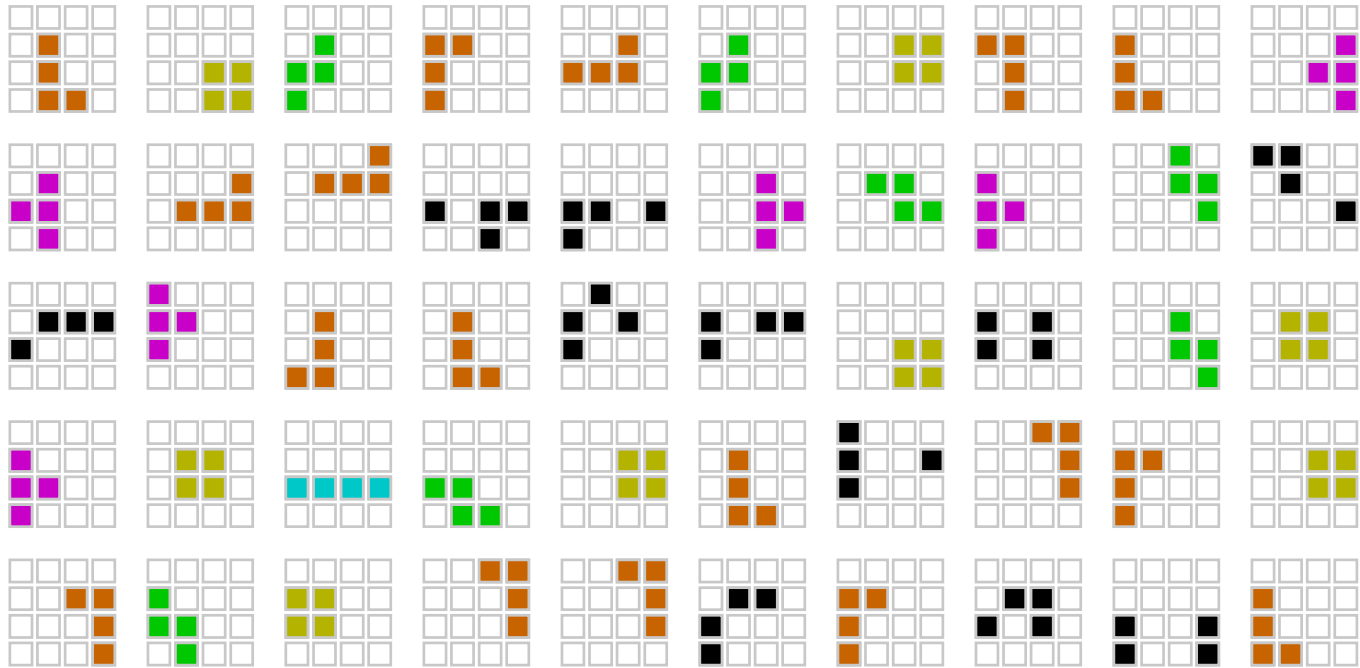
Generation 2



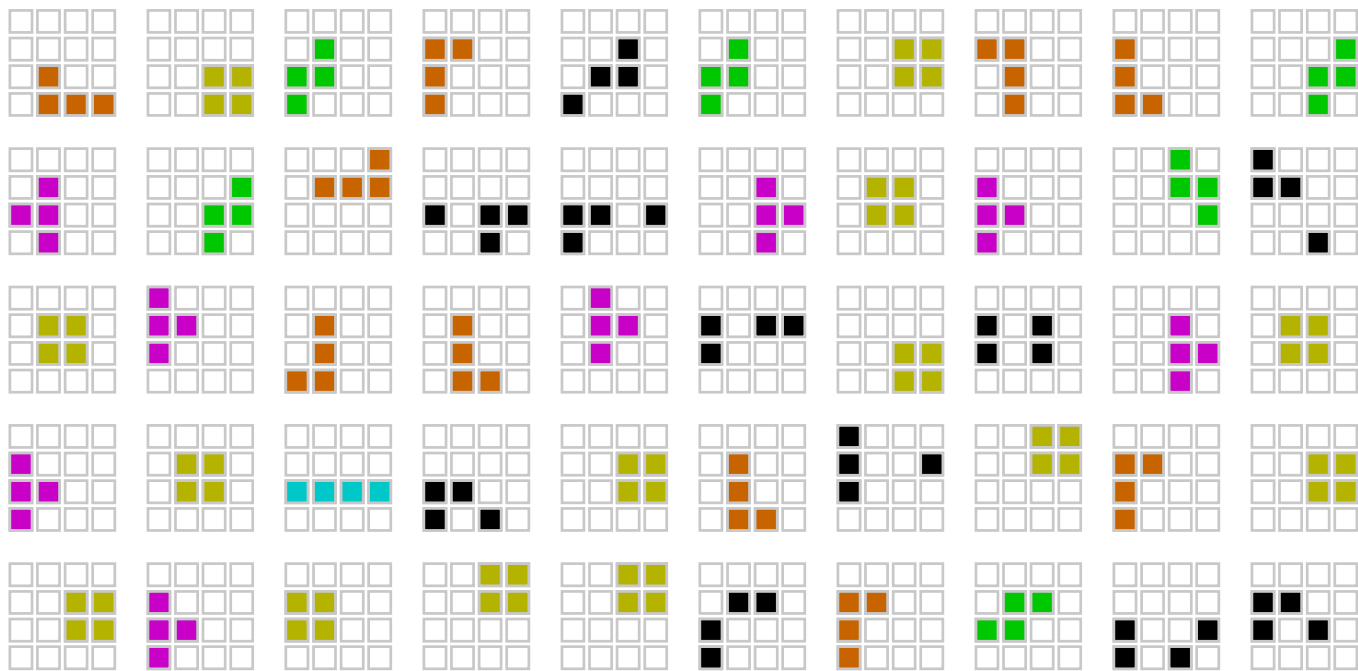
Generation 3



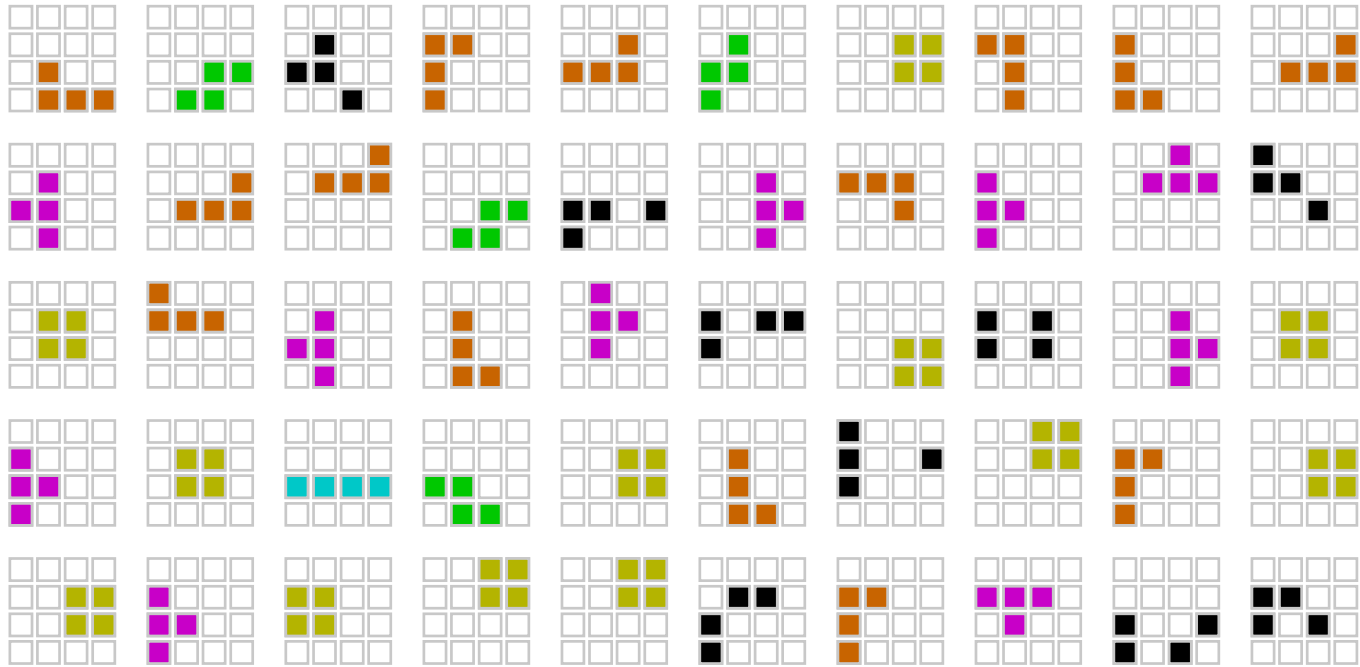
Generation 4



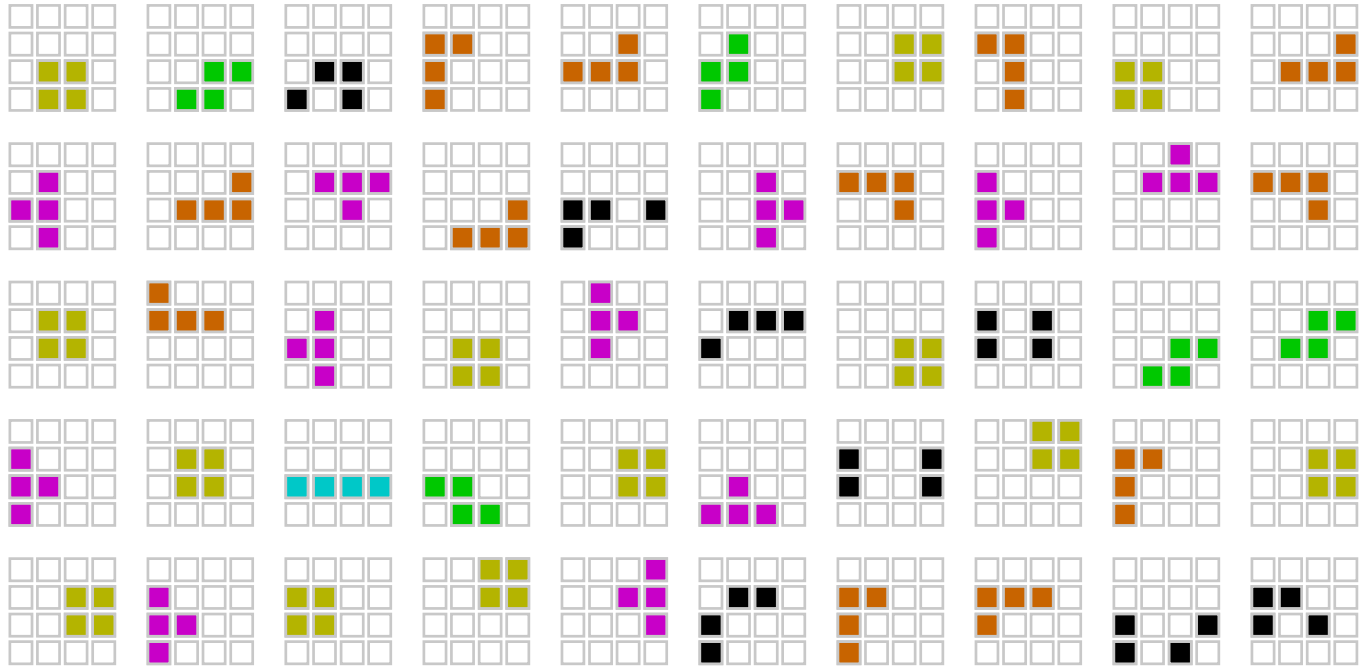
Generation 5



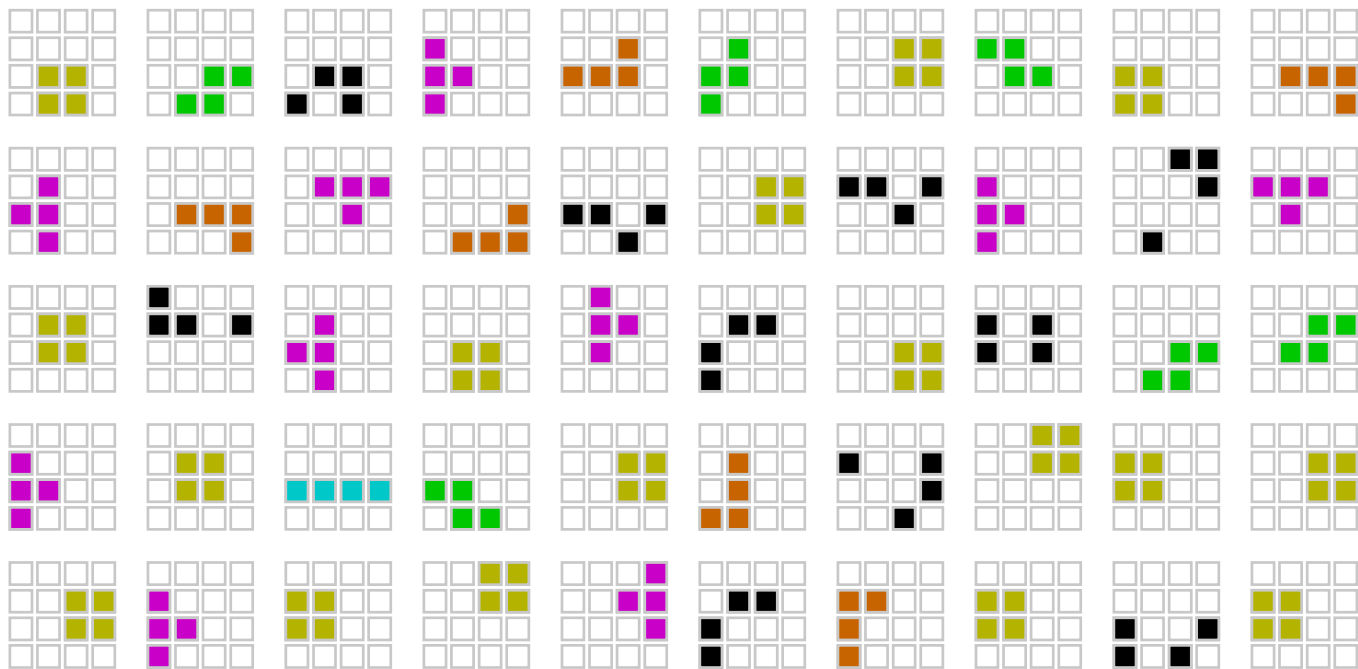
Generation 6



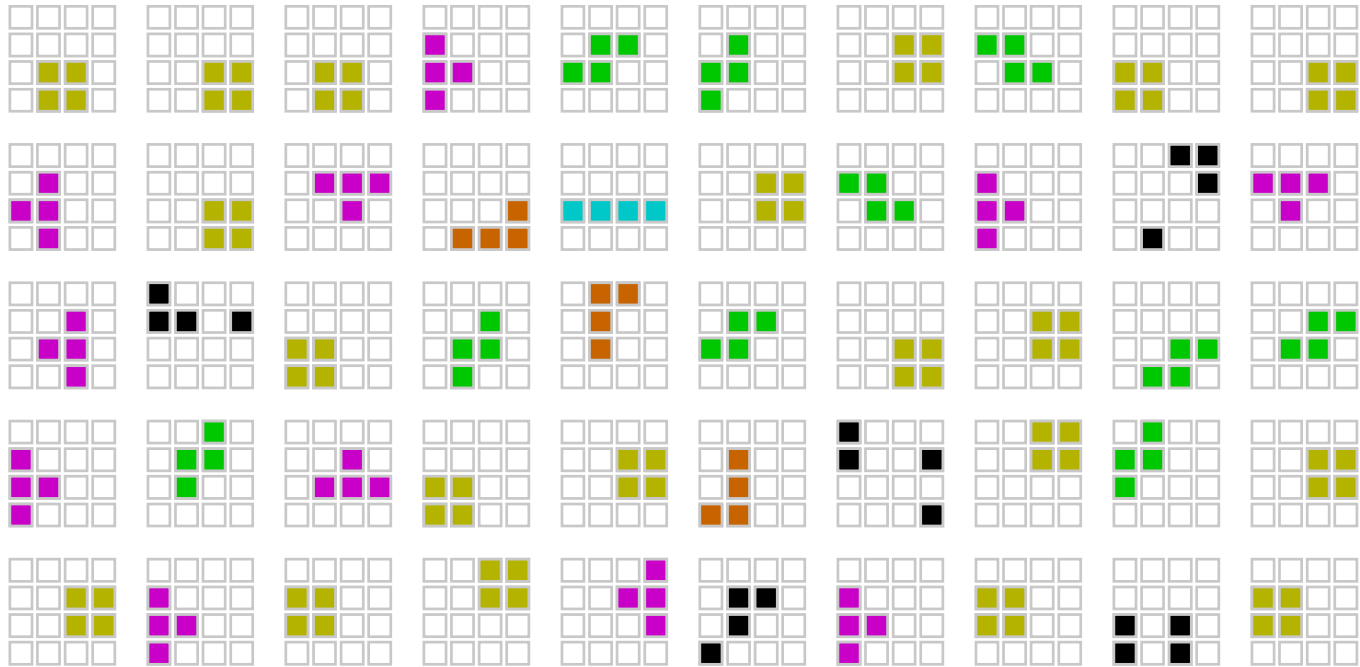
Generation 7



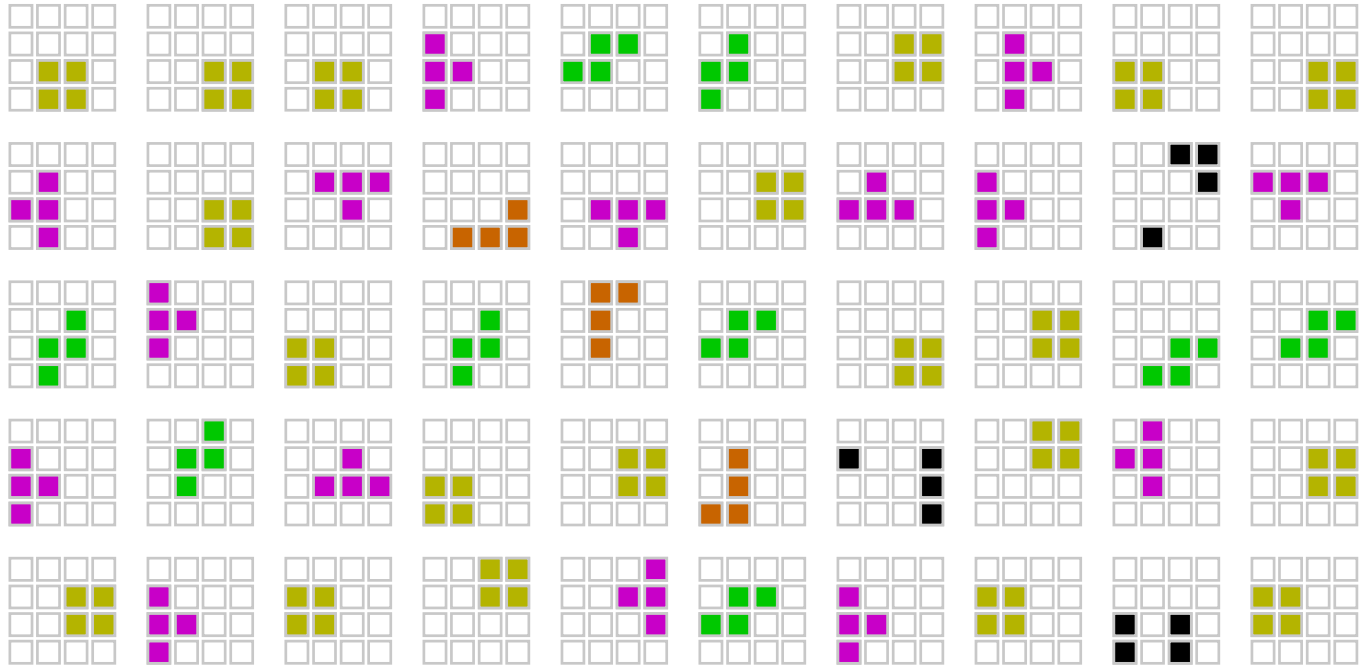
Generation 8



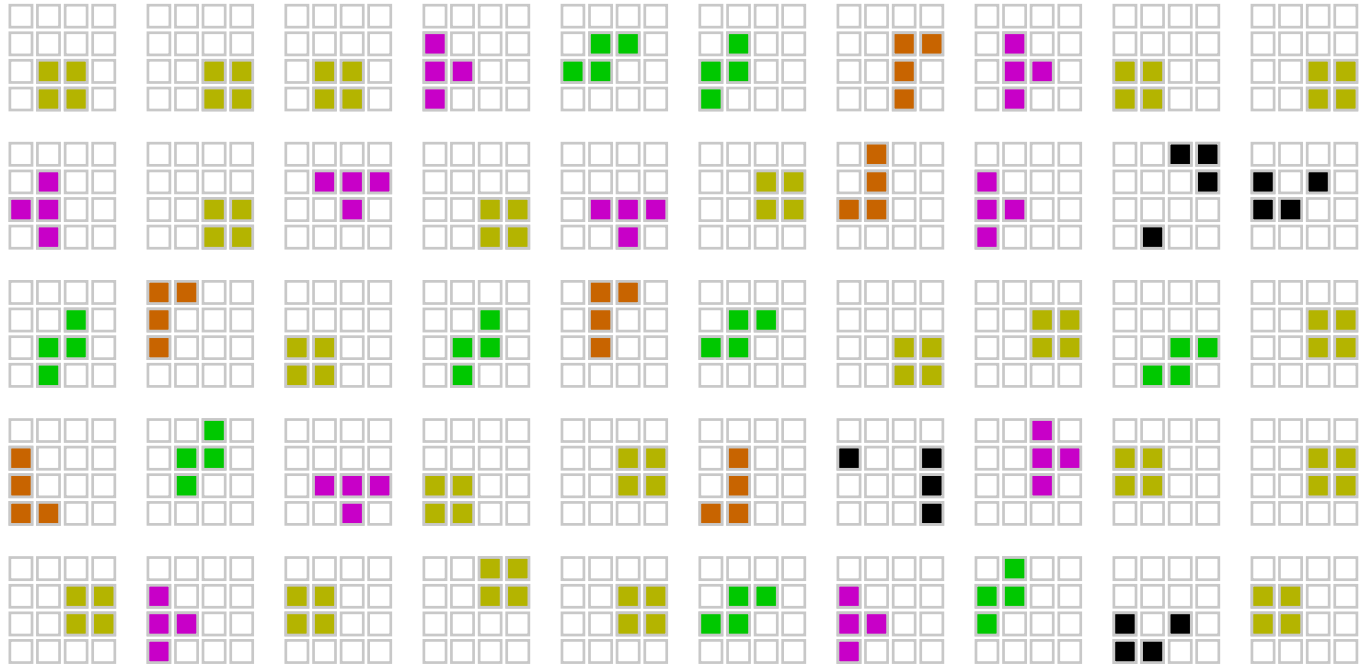
Generation 9



Generation 10



Generation 11



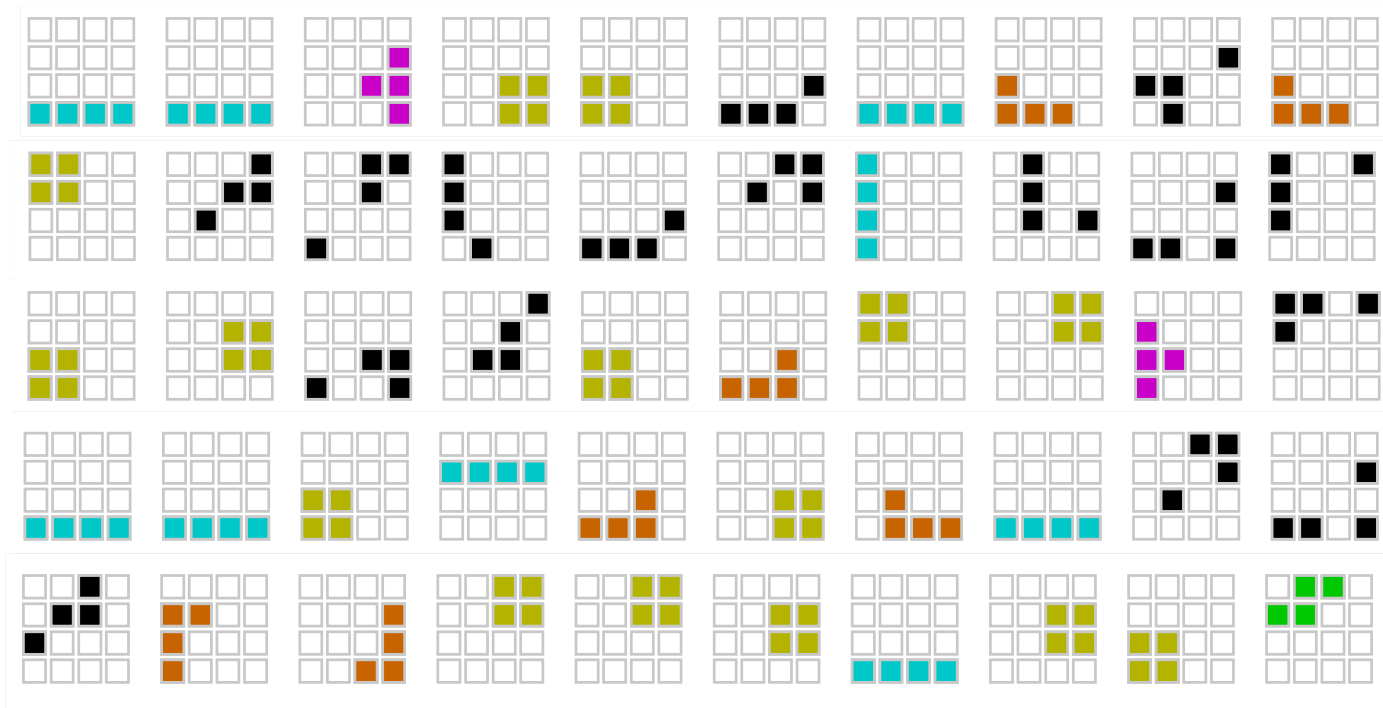
Generation 12

Emergence of a **system**



Chain 4, Generation 12

Emergence of a **system**



Chain 1, Generation 12

Systematic structure develops even
in baboons (if you scaffold their
environment in the right way)

The idea

- Humans ended up with an unusual combination of traits: ubiquitous social learning (including of vocal signalling) and mental interpenetration
- This set in place a cultural evolutionary process that shaped how language works

Schedule

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Readings and next lecture

- What is evolution? What is adaptation? How can we learn about humans by studying other animals?
- Language as a biological adaptation, evolved through natural selection under pressure for communication

